Popper, 1934: conjecture and refutation



Karl Popper, about 1980.

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Karl Popper, *The Logic of Scientific Discovery*, chapter 10, section 85 (1959), the authorised English of *Logik der Forschung* (1934).

Popper, 1934: conjecture and refutation

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Vienna, 1934. Popper asks what entitles anyone to believe a general rule drawn from finitely many cases.

His answer is that nothing does, and that this is survivable.

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We do not know: we can only guess.

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Popper's tetradic schema

The way in which knowledge progresses … is by unjustified (and unjustifiable) anticipations, by guesses, by tentative solutions to our problems, by conjectures. These conjectures are controlled by criticism; that is, by attempted refutations …

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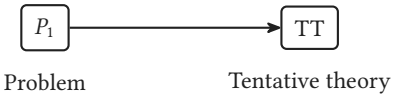


Problem

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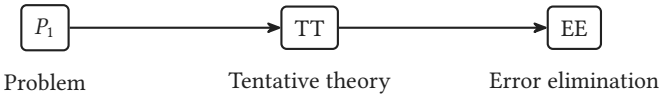
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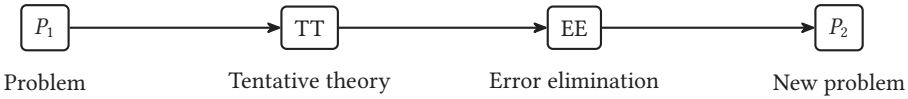
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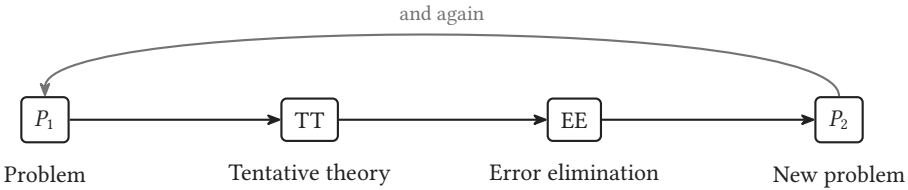
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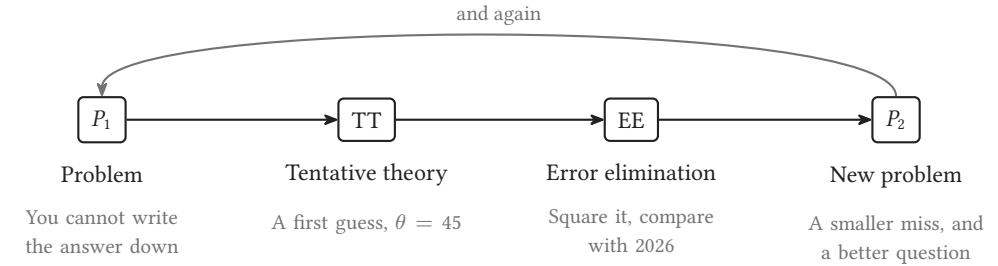
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Turing, 1950: the child machine



Alan Turing, 29 March 1951.

Elliott and Fry

A. M. Turing, “Computing Machinery and Intelligence”, *Mind* LIX(236), 1950, section 7, Learning Machines.

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Elliott and Fry

We cannot expect to find a good child-machine at the first attempt. One must experiment with teaching one such machine and see how well it learns. One can then try another and see if it is better or worse.

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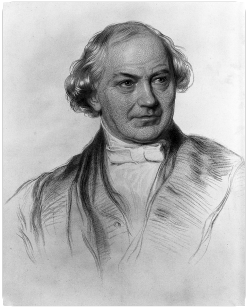
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Now the learning process may be regarded as a search for a form of behaviour which will satisfy the teacher (or some other criterion).

You do not get it right the first time. You try, you measure, you try again.

A. M. Turing, "Computing Machinery and Intelligence", *Mind* LIX(236), 1950, section 7, Learning Machines.

Whewell, 1858: trying wrong guesses



William Whewell, stipple engraving.

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William Whewell, *Novum Organon Renovatum*, third edition (1858), book II, chapter V, page 79.

Whewell, 1858: trying wrong guesses



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Whewell, 1858: trying wrong guesses



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The tendencies of our speculative nature … perpetually show their vigour by overshooting the mark. They obtain something, by aiming at much more.

Most guesses are wrong, and a step that aims too far is how the right one gets found.

William Whewell, *Novum Organon Renovatum*, third edition (1858), book II, chapter V, page 79.

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